

GUAIZIHU, China

WMO: 523780

Lat: 41.367N

Lon: 102.367E

Elev: 3150

StdP: 13.10

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.3	(c) -3.1	(d) -26.4	(e) 1.4	(f) 11.2	(g) -22.5	(h) 1.8	(i) 12.8	(j) 27.7	(k) 18.4	(l) 25.1	(m) 19.1	(n) 4.1	(o) 90	(p) 0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	100.5	62.7	97.6	61.7	95.0	60.8	67.7	88.0	65.7	85.9	64.1	84.8	12.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	94.4	72.8	59.5	85.4	73.3	56.5	76.4	74.0	34.1	87.9	32.4	85.9	31.0	84.7	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.6	25.9	23.0	-11.9	105.0	4.6	2.9	-15.2	107.0	-17.9	108.7	-20.4	110.3	-23.8	112.4
			-12.6	70.2	4.3	3.7	-15.8	72.8	-18.3	75.0	-20.7	77.0	-23.9	79.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	13.8	23.9	38.6	56.0	67.5	78.2	83.0	79.0	67.4	51.0	33.3	17.7
	DBStd	25.17	8.14	9.07	10.23	9.24	7.83	5.95	5.77	5.86	7.56	8.45	8.97	8.72
	HDD50	3865	1121	732	371	45	1	0	0	0	1	91	502	1001
	HDD65	6830	1586	1152	818	292	65	1	0	1	60	439	950	1466
	CDD50	4200	0	0	18	225	544	846	1022	899	523	121	2	0
	CDD65	1690	0	0	0	22	143	397	557	435	132	4	0	0
	CDH74	21408	0	0	1	269	1732	5116	7675	5280	1292	43	0	0
	CDH80	11002	0	0	0	65	668	2644	4452	2717	452	4	0	0
Wind	WSAvg	10.7	8.7	9.7	11.6	12.7	12.4	11.9	11.4	10.8	10.0	9.8	10.3	9.3
Precipitation	PrecAvg	1.3	0.0	0.0	0.0	0.0	0.2	0.2	0.4	0.4	0.2	0.1	0.0	0.0
	PrecMax	3.1	0.1	0.0	0.2	0.6	1.2	1.4	1.8	1.1	0.7	0.9	0.2	0.2
	PrecMin	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.6	0.0	0.0	0.1	0.1	0.2	0.3	0.4	0.3	0.2	0.2	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	51.1	70.7	86.3	95.4	101.3	105.5	101.7	93.9	79.0	60.8	42.7
		MCWB	28.9	35.9	45.4	53.2	56.9	60.9	65.8	64.1	59.5	52.1	42.3	30.8
	2%	DB	33.6	46.0	65.0	80.9	90.9	97.7	101.8	97.9	88.8	73.7	55.7	37.7
		MCWB	25.2	32.5	42.7	50.6	55.7	59.8	63.7	62.6	57.8	49.4	39.3	27.8
	5%	DB	30.1	42.4	60.0	76.3	87.1	95.1	99.1	94.7	85.1	69.6	51.7	33.9
		MCWB	23.0	30.6	40.1	48.7	54.1	59.0	62.7	61.5	56.2	47.5	37.3	25.6
	10%	DB	26.8	38.8	55.3	72.2	83.4	92.3	96.3	92.1	81.4	65.5	47.6	30.7
		MCWB	20.7	28.4	37.8	46.9	52.8	58.1	62.1	61.0	55.1	45.4	35.1	23.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.0	36.5	45.8	54.5	59.7	65.3	71.8	70.4	64.8	54.1	42.6	31.1
		MCDB	38.4	49.0	67.8	80.8	86.1	80.8	89.6	84.7	79.7	72.1	58.0	42.0
	2%	WB	25.7	33.2	43.1	51.8	57.2	63.3	68.5	67.5	61.4	50.6	39.9	28.0
		MCDB	33.3	44.9	62.3	77.5	86.0	83.2	91.0	87.0	78.8	70.7	54.8	36.8
	5%	WB	23.1	30.9	40.9	49.7	55.3	61.8	66.3	65.6	58.7	48.3	37.5	25.9
		MCDB	29.7	41.9	58.4	74.1	82.8	85.8	87.6	84.8	78.3	67.5	51.3	33.7
	10%	WB	20.7	28.4	38.3	47.5	53.8	60.4	64.7	63.7	56.7	46.1	35.1	23.7
		MCDB	26.6	38.4	54.6	71.1	80.5	86.4	86.3	84.1	77.1	63.8	47.2	30.4
Mean Daily Temperature Range	5% DB	MDBR	20.9	23.7	24.5	23.3	23.2	22.3	21.3	21.7	22.1	22.9	20.7	19.1
		MCDBR	22.3	26.6	29.0	27.5	27.6	25.5	24.2	24.4	25.2	26.7	24.1	19.5
		MCWBR	16.3	17.2	16.2	13.2	12.2	9.3	7.6	8.2	9.5	13.4	14.1	13.6
	5% WB	MCDBR	22.1	25.7	26.9	25.3	24.5	21.0	19.7	20.8	21.9	25.3	23.9	19.3
		MCWBR	16.1	16.7	15.2	12.5	11.3	7.5	6.8	7.6	8.5	13.0	14.2	13.6
Clear-Sky Solar Irradiance	taub	0.317	0.372	0.417	0.464	0.479	0.494	0.486	0.466	0.431	0.381	0.341	0.318	
	taud	2.139	2.012	1.914	1.842	1.863	1.892	1.963	2.001	2.027	2.082	2.122	2.126	
	Ebn at Noon	256	256	259	256	255	251	252	253	252	250	243	243	
	Edh at Noon	33	44	55	63	63	62	57	53	48	40	33	31	
All-Sky Solar Radiation	RadAvg	819	1161	1577	1983	2261	2271	2109	1958	1665	1315	915	731	
	RadStd	57	59	122	89	104	127	111	102	92	40	43	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+2.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	+79

Nomenclature: See separate page